

UNIX - ENVIRONMENT

<http://www.tutorialspoint.com/unix/unix-environment.htm>

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An important Unix concept is the **environment**, which is defined by environment variables. Some are set by the system, others by you, yet others by the shell, or any program that loads another program.

A variable is a character string to which we assign a value. The value assigned could be a number, text, filename, device, or any other type of data.

For example, first we set a variables TEST and then we access its value using **echo** command:

```
$TEST="Unix Programming"
$echo $TEST
Unix Programming
```

Note that environment variables are set without using *signbutwhileaccessingthemweuse* sign as prefix. These variables retain their values until we come out shell.

When you login to the system, the shell undergoes a phase called initialization to set up various environment. This is usually a two step process that involves the shell reading the following files –

- /etc/profile
- profile

The process is as follows –

- The shell checks to see whether the file **/etc/profile** exists.
- If it exists, the shell reads it. Otherwise, this file is skipped. No error message is displayed.
- The shell checks to see whether the file **.profile** exists in your home directory. Your home directory is the directory that you start out in after you log in.
- If it exists, the shell reads it; otherwise, the shell skips it. No error message is displayed.

As soon as both of these files have been read, the shell displays a prompt –

```
$
```

This is the prompt where you can enter commands in order to have them execute.

Note – The shell initialization process detailed here applies to all **Bourne** type shells, but some additional files are used by **bash** and **ksh**.

The .profile File

The file **/etc/profile** is maintained by the system administrator of your UNIX machine and contains shell initialization information required by all users on a system.

The file **.profile** is under your control. You can add as much shell customization information as you want to this file. The minimum set of information that you need to configure includes

- The type of terminal you are using
- A list of directories in which to locate commands
- A list of variables effecting look and feel of your terminal.

You can check your **.profile** available in your home directory. Open it using **vi** editor and check all the variables set for your environment.

Setting the Terminal Type

Usually the type of terminal you are using is automatically configured by either the **login** or **getty** programs. Sometimes, the autoconfiguration process guesses your terminal incorrectly.

If your terminal is set incorrectly, the output of commands might look strange, or you might not be able to interact with the shell properly.

To make sure that this is not the case, most users set their terminal to the lowest common denominator as follows –

```
$TERM=vt100
$
```

Setting the PATH

When you type any command on command prompt, the shell has to locate the command before it can be executed.

The PATH variable specifies the locations in which the shell should look for commands. Usually it is set as follows –

```
$PATH=/bin:/usr/bin
$
```

Here each of the individual entries separated by the colon character, :, are directories. If you request the shell to execute a command and it cannot find it in any of the directories given in the PATH variable, a message similar to the following appears –

```
$hello
hello: not found
$
```

There are variables like PS1 and PS2 which are discussed in the next section.

PS1 and PS2 Variables

The characters that the shell displays as your command prompt are stored in the variable PS1. You can change this variable to be anything you want. As soon as you change it, it'll be used by the shell from that point on.

For example, if you issued the command –

```
$PS1='=>'
=>
=>
=>
```

Your prompt would become =>. To set the value of PS1 so that it shows the working directory, issue the command –

```
=>PS1="[u@\h \w]\$"
[root@ip-72-167-112-17 /var/www/tutorialspoint/unix]$
[root@ip-72-167-112-17 /var/www/tutorialspoint/unix]$
```

The result of this command is that the prompt displays the user's username, the machine's name *hostname*, and the working directory.

There are quite a few escape sequences that can be used as value arguments for PS1; try to limit yourself to the most critical so that the prompt does not overwhelm you with information.

Escape Sequence	Description
\t	Current time, expressed as HH:MM:SS.

\d	Current date, expressed as Weekday Month Date
\n	Newline.
\s	Current shell environment.
\W	Working directory.
\w	Full path of the working directory.
\u	Current user.s username.
\h	Hostname of the current machine.
\#	Command number of the current command. Increases with each new command entered.
\\$	If the effective UID is 0 <i>that is, if you are logged in as root</i> , end the prompt with the # character; otherwise, use the \$.

You can make the change yourself every time you log in, or you can have the change made automatically in PS1 by adding it to your **.profile** file.

When you issue a command that is incomplete, the shell will display a secondary prompt and wait for you to complete the command and hit Enter again.

The default secondary prompt is `>` *the greater than sign*, but can be changed by re-defining the **PS2** shell variable –

Following is the example which uses the default secondary prompt –

```
$ echo "this is a
> test"
this is a
test
$
```

Following is the example which re-define PS2 with a customized prompt –

```
$ PS2="secondary prompt->"
$ echo "this is a
secondary prompt->test"
this is a
test
$
```

Environment Variables

Following is the partial list of important environment variables. These variables would be set and accessed as mentioned above –

Variable	Description
DISPLAY	Contains the identifier for the display that X11 programs should use by default.
HOME	Indicates the home directory of the current user: the default argument for the cd built-in command.
IFS	Indicates the Internal Field Separator that is used by the parser for word splitting after expansion.
LANG	LANG expands to the default system locale; LC_ALL can be used to override this. For example, if its value is pt_BR, then the language is set

to *Brazilian* Portuguese and the locale to Brazil.

LD_LIBRARY_PATH	On many Unix systems with a dynamic linker, contains a colon-separated list of directories that the dynamic linker should search for shared objects when building a process image after exec, before searching in any other directories.
PATH	Indicates search path for commands. It is a colon-separated list of directories in which the shell looks for commands.
PWD	Indicates the current working directory as set by the cd command.
RANDOM	Generates a random integer between 0 and 32,767 each time it is referenced.
SHLVL	Increments by one each time an instance of bash is started. This variable is useful for determining whether the built-in exit command ends the current session.
TERM	Refers to the display type
TZ	Refers to Time zone. It can take values like GMT, AST, etc.
UID	Expands to the numeric user ID of the current user, initialized at shell startup.

Following is the sample example showing few environment variables –

```
$ echo $HOME
/root
]$ echo $DISPLAY

$ echo $TERM
xterm
$ echo $PATH
/usr/local/bin:/bin:/usr/bin:/home/amrood/bin:/usr/local/bin
$
```

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